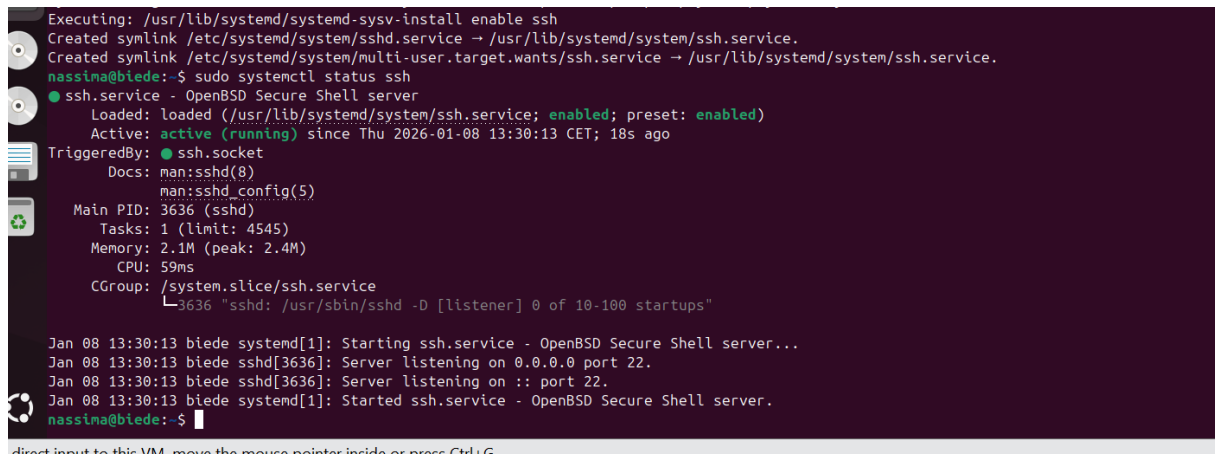


Template Week 6 – Networking

Student number:556707

Assignment 6.1: Working from home

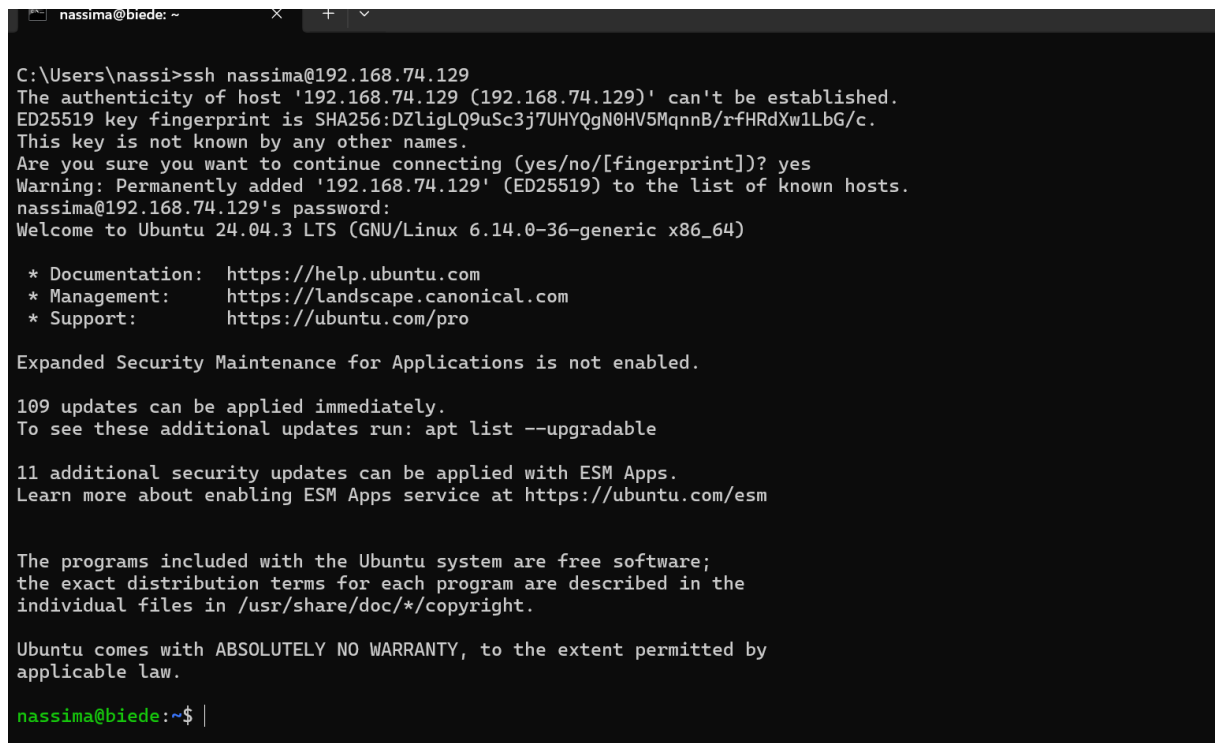
Screenshot installation openssh-server:



```
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
nassima@biede:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-01-08 13:30:13 CET; 18s ago
     TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
            man:sshd_config(5)
    Main PID: 3636 (sshd)
      Tasks: 1 (limit: 4545)
     Memory: 2.1M (peak: 2.4M)
        CPU: 59ms
    CGroup: /system.slice/ssh.service
            └─3636 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jan 08 13:30:13 biede systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jan 08 13:30:13 biede sshd[3636]: Server listening on 0.0.0.0 port 22.
Jan 08 13:30:13 biede sshd[3636]: Server listening on :: port 22.
Jan 08 13:30:13 biede systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
nassima@biede:~$
```

Screenshot successful SSH command execution:



```
C:\Users\nassi>ssh nassima@192.168.74.129
The authenticity of host '192.168.74.129 (192.168.74.129)' can't be established.
ED25519 key fingerprint is SHA256:DZligLQ9uSc3j7UHYQgN0HV5MqnnB/rfHRdXw1LbG/c.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.74.129' (ED25519) to the list of known hosts.
nassima@192.168.74.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-36-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

109 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

11 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM service at https://ubuntu.com/esm

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

nassima@biede:~$
```

Screenshot successful execution SCP command:

```
scp_test.txt                                100%  19    4.4KB/s   00:00
nassima@biede:~$ exit
logout
Connection to 192.168.74.129 closed.

C:\Users\nassi>echo Hello from Windows > scp_test.txt

C:\Users\nassi>scp scp_test.txt nassima@192.168.74.129:/home/nassima/
nassima@192.168.74.129's password:
scp_test.txt                                100%  21    2.9KB/s   00:00

C:\Users\nassi>
```

Screenshot remmina:

Assignment 6.2: IP addresses websites

Relevant screenshots nslookup command:

Screenshot website visit via IP address:

Assignment 6.3: subnetting

How many IP addresses are in this network configuration 192.168.110.128/25?

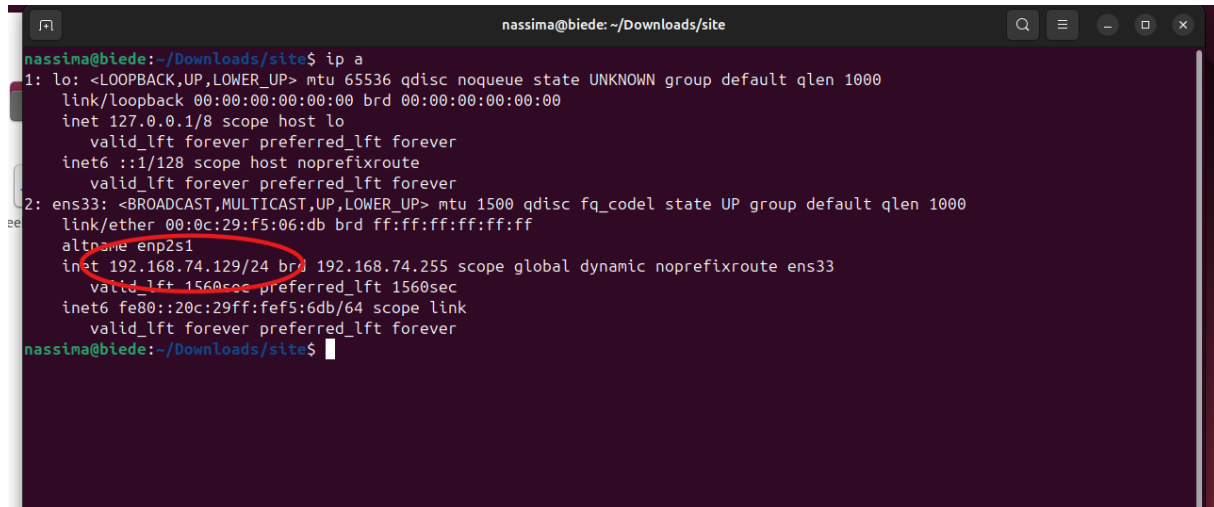
What is the usable IP range to hand out to the connected computers?

Check your two previous answers with this Linux command: `ipcalc 192.168.110.128/25`

Explain the above calculation in your own words.

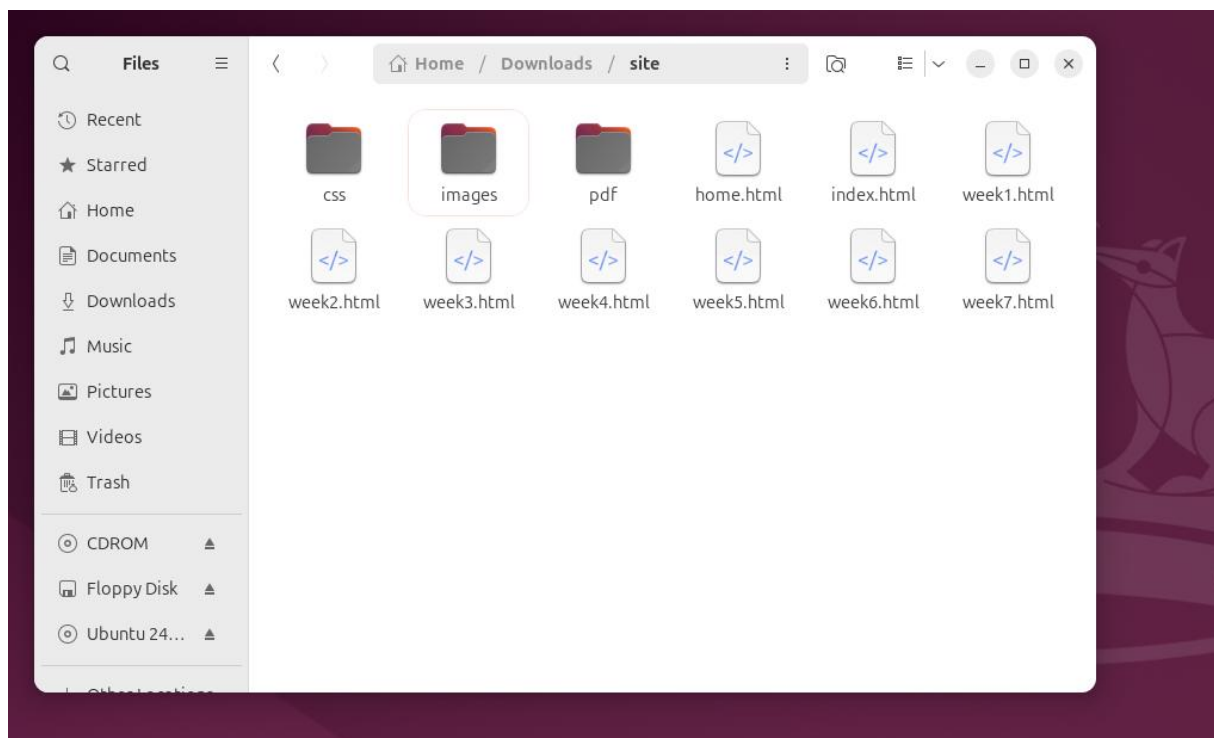
Assignment 6.4: HTML

Screenshot IP address Ubuntu VM:



```
nassima@biede: ~/Downloads/site$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f5:06:db brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.74.129/24 brd 192.168.74.255 scope global dynamic noprefixroute ens33
        valid_lft 1560sec preferred_lft 1560sec
    inet6 fe80::20c:29ff:fef5:6db/64 scope link
        valid_lft forever preferred_lft forever
nassima@biede: ~/Downloads/site$
```

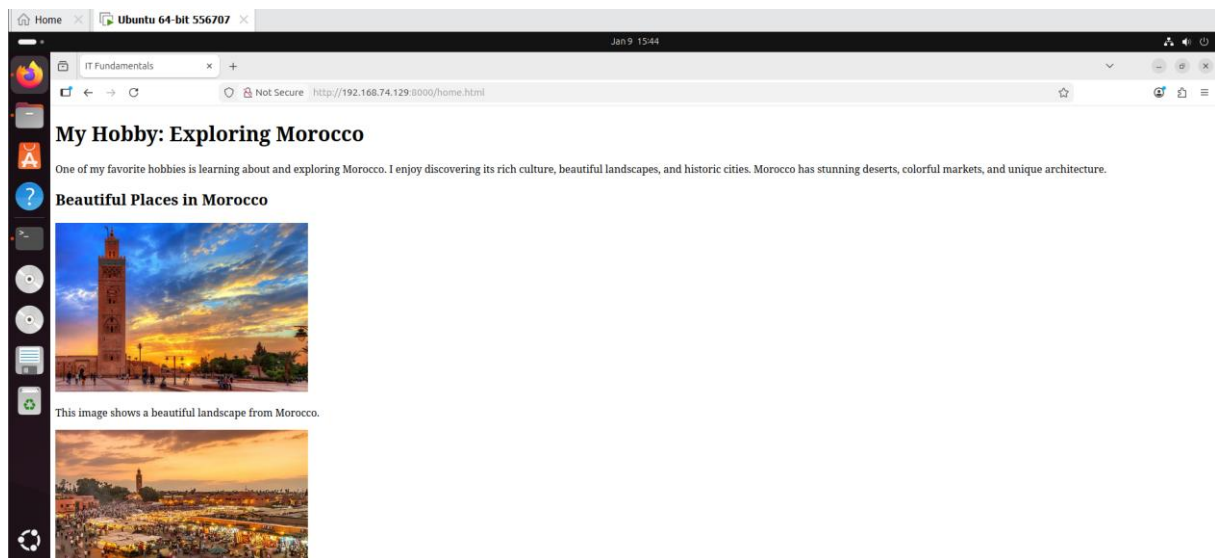
Screenshot of Site directory contents:



Screenshot python3 webserver command:

```
nassima@biede: ~/Downloads/site
nassima@biede:~/Downloads/site$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

Screenshot web browser visits your site



Assignment 6.5: Network segment

Remember that bitwise java application you've made in week 2? Expand that application so that you can also calculate a network segment as explained in the PowerPoint slides of week 6. Use the bitwise & AND operator. You need to be able to input two Strings. An IP address and a subnet.

IP: 192.168.1.100 and subnet: 255.255.255.224 for /27

Example: 192.168.1.100/27

Calculate the network segment

IP Address: 11000000.10101000.00000001.01100100

Subnet Mask: 11111111.11111111.11111111.11100000

Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.
For a /27 subnet, each segment (or subnet) has 32 IP addresses (2^5).
The range of this network segment is from 192.168.1.96 to 192.168.1.127.

Paste source code here, with a screenshot of a working application.

Ready? Save this file and export it as a pdf file with the name: [week6.pdf](#)